1 Experimental Specification for the White-Cat, Black-Cat, and Gray-Cat Metastable Interface in AB-C-D Stages v1

Subtitle: Closed-system numerical experiment on the AB two-body metastable interface, weak C readout, and strong D observation for white-cat, black-cat, and gray-cat branching
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Series: Wave Information Readout / staged experiment for macroscopic cat-like metastable states
Target implementation: Python numerical experiment
Version DOI: 10.5281/zenodo.21353209
Concept DOI: 10.5281/zenodo.21353208

1.1 0. Summary

This experiment does not assume in advance that an equal-allocation state of the white-cat state `A` and the black-cat state `B` is merely an unselected superposition.

First, the AB two-body system alone is used to distinguish three phases.

```
gray eigen phase:
  A/B remains fixed near 0.5/0.5.

gray metastable phase:
  A/B oscillates weakly near 0.5/0.5 but does not naturally converge to one side.

natural selection phase:
  A or B is selected even without C or D.
```

Then C is added to test whether the metastable state can be read without selection. Finally D is added to test whether observation selects either A or B.

The first checkpoint is not the action of C or D.

```
The first checkpoint is to find the interface between the gray eigen phase,
gray metastable phase, and natural selection phase in AB alone.
```

1.2 1. Purpose

The experiment checks the following stages in order.

1. Classify the gray eigen phase, gray metastable phase, and natural selection phase in AB alone.
2. Add C and test whether a gray eigen or metastable state can be read without selection.
3. Add D and test the conditions under which observation selects A or B.
4. Confirm whether a gray eigen phase remains gray even under D.
5. In large-amplitude regions where A/B is already separated, confirm that D agrees with the C readout.

D is not assumed to always cause collapse.

If a gray cat has already become an eigen state, it does not need to fall into white or black under D.

If A/B is already clearly separated, the D result is treated as a readout of an already existing A/B branching, not as a selection created by D.

1.3 2. State Variables

At minimum, the following variables are recorded.

variable	meaning
`a`	complex amplitude of white-cat A
`b`	complex amplitude of black-cat B
`p_A = |a|^2`	white-cat allocation
`p_B = |b|^2`	black-cat allocation
`Q = p_A + p_B`	total A/B amount
`S = p_A − p_B`	A/B selection order variable
`S_amp`	oscillation amplitude of `S`
`S_drift`	long-time drift of `S`
`C_A, C_B`	A/B readout by C
`D_A, D_B`	A/B readout by D
`L_A, L_B`	localization of A/B
`N_eff_A, N_eff_B`	effective harmonic order of A/B
`E_total`	total conserved quantity or conservation registry quantity

The central checks are:

```
whether Q is conserved;
whether S is fixed, weakly oscillating, or naturally growing;
whether C and D change the phase classification of S.
```

1.4 3. AB Two-Body Metastable Interface Experiment

1.5 3.1 Purpose

The AB system alone is used to test what kind of state the gray cat is.

```
C = off
D = off
```

No observation-induced selection is included at this stage.

1.6 3.2 Initial State

The reference initial values are

$$a_0 = \frac{1}{\sqrt{2}},$$

$$b_0 = \frac{e^{i\phi_0}}{\sqrt{2}}.$$

Then

$$Q_0 = |a_0|^2 + |b_0|^2 = 1,$$

$$S_0 = |a_0|^2 - |b_0|^2 = 0.$$

When a small initial bias is inserted,

$$p_{A,0} = \frac{1}{2} + s_0,$$

$$p_{B,0} = \frac{1}{2} - s_0.$$

1.7 3.3 AB Exchange Map

The minimal AB exchange map is

$$\begin{pmatrix} a_{k+1} \\ b_{k+1} \end{pmatrix} = U_\epsilon \begin{pmatrix} a_k \\ b_k \end{pmatrix},$$

$$U_\epsilon = \begin{pmatrix} \cos\epsilon & i\sin\epsilon \\ i\sin\epsilon & \cos\epsilon \end{pmatrix}.$$

This map alone gives a simple unitary exchange oscillation.

If needed, a weak restoring term or weak nonlinear term is added to test closed-system stability. However, a term that causes one-sided natural convergence without C or D is not used for the cat-selection experiment.

1.8 3.4 Sweep Parameters

At minimum, the following parameters are swept.

parameter	meaning
`epsilon`	strength of AB exchange oscillation
`phi_0`	initial relative phase
`s_0`	small initial A/B bias
`noise_amp`	numerical fluctuation or small disturbance
`K`	number of evolution steps

When the gray cat is treated as a harmonic metastable state, the amplitude of S is kept small. The target condition is

```
S_amp < S_gray_limit.
```

The initial guide values are:

```
S_gray_limit = 0.05
S_amp target = about 0.01 to 0.03
```

1.9 3.5 Phase Classification

The AB two-body results are classified into three phases.

1.9.1 Gray Eigen Phase

```
S_mean ≈ 0
S_amp -> 0
S_drift ≈ 0
Q ≈ 1
```

In this case, the equal A/B allocation is treated as a new gray-cat eigen state.

1.9.2 Gray Metastable Phase

```
S_mean ≈ 0
0 < S_amp < S_gray_limit
S_drift ≈ 0
Q ≈ 1
```

This phase is the main target for the D selection experiment.

1.9.3 Natural Selection Phase

```
abs(S) grows without C or D
S -> +1 or S -> -1
```

In this phase, observation-induced selection cannot be claimed.

1.10 4. C Readout Experiment

1.11 4.1 Purpose

C is added to a gray eigen or gray metastable state found in AB.

```
C = on
D = off
```

C reads the A/B allocation. At this stage, C must not push the system into one side.

1.12 4.2 C Readout Quantities

C reads:

$$C_A \approx p_A,$$

$$C_B \approx p_B.$$

The error is recorded as

$$E_C = |C_A - p_A| + |C_B - p_B|.$$

1.13 4.3 C Acceptance Conditions

For the gray metastable phase:

```
E_C < tol_C
abs(S_afterC - S_beforeC) < epsilon_C
no selection
```

For the gray eigen phase:

```
C_A ≈ 0.5
C_B ≈ 0.5
S remains near 0
```

For the large-amplitude region, C reads the current A/B bias.

```
C_A > C_B if S > 0
C_B > C_A if S < 0
```

1.14 5. D Observation Experiment

1.15 5.1 Purpose

D is then added.

```
C = optional
D = on
```

D is a strong observation map used to test whether the system falls into either A or B.

The result is classified by state phase.

1.16 5.2 D in the Gray Eigen Phase

For the gray eigen phase, the test asks whether the gray state remains gray under D.

The expected judgment is:

```
D_A ≈ 0.5
D_B ≈ 0.5
S remains near 0
```

In this case, not falling into white or black is the result.

1.17 5.3 D in the Gray Metastable Phase

For the gray metastable phase, the test asks whether D selects one side.

The judgment is:

```
D_A ≈ 1, D_B ≈ 0
```

or

```
D_A ≈ 0, D_B ≈ 1
```

If the same fall occurs without D, it is not judged as D-induced selection.

1.18 5.4 D in the Large-Amplitude Separated Region

When A/B is already clearly separated, D only needs to read the current A/B bias.

The D readout should agree with the C readout:

```
sign(D_A − D_B) = sign(C_A − C_B)
```

If this agreement is obtained, D is judged to have read an already existing A/B bias, rather than creating the selection.

1.19 6. Four-Stage Main Experiment

1.20 Stage 1: AB Metastable Interface Search

```
C = off
D = off
```

Output:

```
gray eigen phase
gray metastable phase
natural selection phase
```

1.21 Stage 2: C Readout Confirmation

```
C = on
D = off
```

Output:

```
C_read_error
C_induced_bias
C_selection_triggered
```

1.22 Stage 3: D Observation Confirmation

```
C = off or recorded
D = on
```

Output:

```
D_result
D_induced_selection
D_vs_C_agreement
```

1.23 Stage 4: C-D Comparison

The same AB initial condition is compared under C readout and D observation.

Checks:

```
gray eigen phase:
  C and D both read 0.5/0.5.

gray metastable phase:
  C reads but does not select.
  D is tested for one-sided selection.

large-amplitude separated region:
  C and D read the same A/B bias.
  This is not treated as observation-created selection.
```

1.24 7. Acceptance Conditions

The minimum acceptance conditions are:

1. AB alone can classify the gray eigen phase, gray metastable phase, and natural selection phase.
2. In the gray metastable phase, C can read the A/B allocation without selecting one side.
3. In the gray eigen phase, D does not make the system fall into white or black.
4. In the gray metastable phase, D-induced selection can be judged by comparison with a no-D control.
5. In the large-amplitude separated region, D agrees with the C readout.
6. `Q` or the total conservation registry remains conserved within tolerance.

1.25 8. Failure Conditions

The experiment does not proceed to later C/D stages if:

1. All AB-only conditions naturally fall into selection.
2. `Q` is not conserved.
3. C always causes selection.
4. C cannot read the A/B allocation.
5. D/no-D differences cannot be classified.

1.26 9. Result Fields

1.27 9.1 Stage 1: AB Metastable Interface Search

Output:

```
gray_cat_ab_metastable_interface_preliminary_result_v1/
```

Result:

```
total_cases = 5600
gray_eigen = 1248
gray_metastable = 733
large_oscillation = 470
natural_selection = 1070
unstable_or_drifting = 2079
```

1.28 9.2 Stage 2: C Readout Window

Output:

```
gray_cat_c_readout_window_preliminary_result_v1/
```

Result:

```
total_cases = 1080
C_window_count = 247
C_informative_window_count = 144
C_nonzero_backaction_window_count = 114
```

1.29 9.3 Stage 3: D Observation Response

Output:

```
gray_cat_d_observation_response_preliminary_result_v1/
```

Result:

```
total_cases = 7056
D_induced_selection_count = 2016
gray_kept_eigen = 1062
white_selected = 1244
black_selected = 772
```

1.30 9.4 Stage 4: D Selection Boundary

Output:

```
gray_cat_d_selection_boundary_preliminary_result_v1/
```

Result:

```
boundary_points = 438
selection_possible_boundary_points = 438
no_selection_boundary_points = 0
min_D_gain_overall = 0.0225
max_min_D_gain_overall = 0.065
```